

Quiz 1 – Grading Guide with Solutions

Instructor Name: _____

Course: Probability and Statistics

Quiz: Probability & Bayes' Theorem (Machines)

Problem Setup

Factory produces widgets from three machines: M_1 , M_2 , M_3 .

Some widgets are defective.

Events:

- M_i : Widget comes from Machine i
- D : Widget is defective

Conditional probabilities:

Machine	Conditional Probabilities
M_1	$P(M_1 D) = 0.2, P(D M_1) = 0.01$
M_2	$P(M_2 D) = 0.3, P(D M_2) = 0.03$
M_3	$P(M_3 D) = 0.5, P(D M_3) = 0.05$

Grading and Solutions

Question 1 (4 pts)

Compute $P(D)$

Formula: Law of total probability:

$$P(D) = P(D | M_1)P(M_1) + P(D | M_2)P(M_2) + P(D | M_3)P(M_3)$$

Solution:

We first compute $P(M_i)$ using Bayes' theorem from given $P(M_i | D)$ and $P(D | M_i)$ (see Q2):

- $P(M_1) = \frac{0.2 \cdot P(D)}{0.01} = 20P(D)$
- $P(M_2) = \frac{0.3 \cdot P(D)}{0.03} = 10P(D)$
- $P(M_3) = \frac{0.5 \cdot P(D)}{0.05} = 10P(D)$

Sum of probabilities = 1: $20P(D) + 10P(D) + 10P(D) = 40P(D) = 1$

$$\Rightarrow P(D) = 0.025$$

Answer: $P(D) = 0.025$

Points Allocation:

- 2 pt: Correct formula
 - 2 pt: Correct substitution
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Question 2 (4 pts)Determine $P(M_1), P(M_2), P(M_3)$

Formula: Bayes' theorem:

$$P(M_i) = \frac{P(M_i | D)P(D)}{P(D | M_i)}$$

Solution:

- $P(M_1) = \frac{0.2 \cdot 0.025}{0.01} = 0.5$
- $P(M_2) = \frac{0.3 \cdot 0.025}{0.03} = 0.25$
- $P(M_3) = \frac{0.5 \cdot 0.025}{0.05} = 0.25$

Answer: $P(M_1) = 0.5, P(M_2) = 0.25, P(M_3) = 0.25$ **Points Allocation:**

- 2 pt: Correct formula
 - 2 pt: Correct substitution
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Question 3 (4 pts)Compute $P(M_i \cap D)$

Formula:

$$P(M_i \cap D) = P(D | M_i)P(M_i)$$

Solution:

- $P(M_1 \cap D) = 0.01 \cdot 0.5 = 0.005$
- $P(M_2 \cap D) = 0.03 \cdot 0.25 = 0.0075$
- $P(M_3 \cap D) = 0.05 \cdot 0.25 = 0.0125$

Answer: $P(M_1 \cap D) = 0.005, P(M_2 \cap D) = 0.0075, P(M_3 \cap D) = 0.0125$ **Points Allocation:**

- 2 pt: Correct formula
 - 2 pt: Correct substitution
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Question 4 (4 pts)**Mutual exclusivity and independence**

- **Mutually exclusive?**
 $M_1 \cap D$ and $M_2 \cap D$ cannot occur simultaneously - **Yes**, all pairs are mutually exclusive.
- **Independence check:**
 Check if $P(M_1 \cap D) = P(M_1)P(D)$:
 $0.005 \neq 0.5 \cdot 0.025 = 0.0125$ - **Not independent**
 Similarly, $M_2 \cap D$ and $M_3 \cap D$ are **not independent**.

Points Allocation:

- 1 pt: Correct answer for mutual exclusivity
- 1 pt: Correct answer for independence

- 1 pt: Supporting calculations/formulas
 - 1 pt: Clear explanation
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Question 5 (4 pts)

Two widgets selected with replacement: probability at least one defective

- Probability none defective:

$$P(\text{none defective}) = (1 - P(D))^2 = (1 - 0.025)^2 = 0.975^2 \approx 0.950625$$

- Probability at least one defective:

$$P(\text{at least one defective}) = 1 - 0.950625 \approx 0.049375$$

Answer: $P(\text{at least one defective}) \approx 0.0494$

Points Allocation:

- 2 pt: Correct formula
 - 2 pt: Correct substitution
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